



Warsaw University of Technology

Faculty of Power and Aeronautical Engineering

Institute of Heat Engineering

Comparative Analysis of Rocket Propellant Combustion Performance Using Cantera

An Extended Propellant Analysis Tool with Isp Calculation

Course: Computer Methods in Combustion (MKWS)

Author: Marcin Mokrosiński

Student ID: 327620

Supervisor: dr inż. Mateusz Żbikowski

Warsaw, 2026

Abstract

This report presents a comprehensive comparative analysis of five rocket propellant combinations — LOX/LH₂, LOX/CH₄, LOX/RP-1, N₂O₄/UDMH, and N₂O/C₂H₄ — using the Cantera thermochemical library. The developed tool extends previous work on propellant analysis by implementing a full isentropic nozzle expansion model, enabling the calculation of specific impulse (I_{sp}) previously absent from similar tools. Key parameters including adiabatic flame temperature, characteristic velocity c^* , specific impulse under vacuum and sea-level conditions, and combustion product compositions are computed across the full operational O/F ratio range for each propellant combination. Results are validated against published performance data for operational rocket engines including the SpaceX Raptor, RS-25, and Merlin 1D, with differences of 14–23%, consistent with the known overestimation of ideal equilibrium models which neglect finite-rate kinetics and boundary-layer losses. An interactive Streamlit web dashboard is provided for real-time parametric exploration. All calculations are performed at a nominal chamber pressure of 100 bar with parametric sweeps from 10 to 300 bar.

Contents

1	Introduction	3
2	Literature Review	3
2.1	Rocket propellant thermodynamics	3
2.2	Specific impulse and nozzle performance	4
2.3	Propellant combinations under study	4
3	Model Description	5
3.1	Thermochemical framework	5
3.2	Chamber equilibrium calculation	5
3.3	Nozzle exit conditions and I_{sp} calculation	5
3.4	Parametric study	6
3.5	Software implementation	6
4	Results	7
4.1	Adiabatic flame temperature	7
4.2	Specific impulse	8
4.3	Combustion product composition	9
4.4	Parametric analysis: effect of chamber pressure	9
4.5	Validation against real engines	10
4.6	Density specific impulse	10
4.7	Characteristic velocity	11
4.8	Interactive dashboard	12
5	Conclusions	12

1 Introduction

The selection of propellant combinations is one of the most critical decisions in rocket engine design, directly determining mission performance, system complexity, and safety margins. Specific impulse (I_{sp}) — the ratio of thrust to propellant mass flow rate — is the primary figure of merit, while combustion temperature, density specific impulse, and exhaust composition determine secondary constraints such as thermal management, volumetric efficiency, and environmental impact [1].

Cantera is an open-source software suite for problems involving chemical kinetics, thermodynamics, and transport processes [4]. While widely used in combustion research, its application to rocket propellant performance analysis — particularly incorporating nozzle thermodynamics for I_{sp} calculation — has been limited in academic course projects. Previous work at this institution [7] developed a Propellant Analysis Tool (PAT) in MATLAB using Cantera, calculating chamber equilibrium properties for two propellant combinations but explicitly noting the absence of I_{sp} computation as a limitation. This work directly addresses that gap.

The objectives of this project are:

1. Implement a Python-based propellant analysis tool using Cantera’s Python API with isentropic nozzle expansion for I_{sp} calculation.
2. Perform a systematic comparative analysis of five propellant combinations across operational O/F ratios and chamber pressures.
3. Validate results against published performance data of operating rocket engines.
4. Provide an interactive web-based dashboard for parametric exploration.

2 Literature Review

2.1 Rocket propellant thermodynamics

The theoretical performance of a rocket propellant is determined by the equilibrium state of the combustion products in the chamber and the subsequent isentropic expansion through the nozzle. The adiabatic flame temperature T_{ad} for a given propellant mixture is found by solving the enthalpy balance at constant pressure [2]:

$$H_{reactants}(T_0, p_c) = H_{products}(T_{ad}, p_c) \quad (1)$$

Cantera solves this problem using the `equilibrate('HP')` command, which finds the chemical equilibrium composition and temperature at constant enthalpy and pressure

[5].

2.2 Specific impulse and nozzle performance

The specific impulse is defined as:

$$I_{sp} = \frac{V_e}{g_0} \quad (2)$$

where V_e is the effective exhaust velocity and $g_0 = 9.806\,65 \text{ m/s}^2$ is the standard gravitational acceleration. For an ideal nozzle with isentropic expansion, the exit velocity can be determined from the enthalpy drop between the chamber state and the exit state:

$$V_e = \sqrt{2(h_c - h_e)} \quad (3)$$

where h_c is the specific enthalpy at chamber conditions and h_e is the specific enthalpy at the nozzle exit, obtained after isentropic expansion to the exit pressure p_e . The characteristic velocity is given by:

$$c^* = \frac{\sqrt{\gamma RT_c}}{\Gamma} \quad (4)$$

where $\Gamma = \sqrt{\gamma} \left(\frac{2}{\gamma+1} \right)^{(\gamma+1)/[2(\gamma-1)]}$ is the Vandekerckhove function [3].

2.3 Propellant combinations under study

The five propellant combinations analysed in this work span the range of practical rocket propellants:

- **LOX/LH₂** — highest specific impulse, used in the Space Shuttle Main Engine (SSME/RS-25) and Ariane 5 Vulcain engine. Storage requires cryogenic temperatures for both components [3].
- **LOX/CH₄** — chosen for the SpaceX Raptor engine due to its advantages in propellant production on Mars and compatibility with full-flow staged combustion cycles [8].
- **LOX/RP-1** — kerosene-based propellant used in SpaceX Merlin, RD-180, and the historic F-1 engine of Saturn V [1].
- **N₂O₄/UDMH** — storable hypergolic propellant used in the Soviet/Russian Proton launch vehicle [9].
- **N₂O/C₂H₄** — self-pressurising, commonly used in student and amateur hybrid rocket projects [10].

3 Model Description

3.1 Thermochemical framework

All thermochemical calculations are performed using Cantera 3.x with the GRI-Mech 3.0 reaction mechanism (`gri30.yaml`), which comprises 325 elementary reactions among 53 species [6]. The mechanism was chosen for its comprehensive coverage of hydrogen, methane, ethane, ethylene, and nitrogen oxide chemistry.

3.2 Chamber equilibrium calculation

For each propellant combination at a given O/F ratio and chamber pressure p_c , the chamber state is determined by:

1. Initialising a Cantera `Solution` object with `gri30.yaml`.
2. Setting the mixture composition according to the O/F mass ratio:

$$x_{ox} = \frac{\text{O/F} \cdot M_{fuel}}{M_{ox}} \quad (5)$$

where M_{fuel} and M_{ox} are the molar masses of fuel and oxidizer.

3. Calling `gas.equilibrate('HP')` to find the adiabatic equilibrium state at constant enthalpy and pressure.

The following chamber parameters are extracted: T_c , $\gamma = c_p/c_v$, $\bar{M}$ (mean molecular weight), c^* , sound speed, and mole fractions of the ten most abundant species.

3.3 Nozzle exit conditions and I_{sp} calculation

The isentropic nozzle expansion is modelled by:

1. Recording the chamber enthalpy h_c and entropy s_c after equilibration.
2. Setting the gas state to (s_c, p_e) — the same entropy at exit pressure.
3. Calling `gas.equilibrate('SP')` to obtain the shifted equilibrium composition at exit conditions.
4. Computing the exit velocity from equation (3) and specific impulse from equation (2).

Two exit pressure conditions are considered: $p_{e,vac} = 0.1$ Pa (vacuum, representative of upper-stage operation) and $p_{e,SL} = 101\,325$ Pa (sea level, representative of first-stage operation).

3.4 Parametric study

The following parameter space is explored:

- O/F ratio: specific range for each propellant (see Table 1)
- Chamber pressure: $p_c \in \{10, 30, 50, 100, 200, 300\}$ bar
- Initial temperature: $T_0 = 300$ K for all cases

Table 1: O/F ratio sweep ranges and stoichiometric values for each propellant.

Propellant	Fuel	O/F min	O/F max	Stoich. O/F
LOX/LH ₂	H ₂	2.0	10.0	7.94
LOX/CH ₄	CH ₄	1.5	6.0	3.97
LOX/C ₂ H ₅ OH	C ₂ H ₅ OH	1.0	4.0	2.09
N ₂ O ₄ /UDMH	N ₂ H ₄	0.5	3.0	2.07
N ₂ O/C ₂ H ₄	C ₂ H ₄	2.0	8.0	6.67

[†] C₂H₅OH substituted by C₂H₆ (ethane); N₂H₄ substituted by NH₃ (ammonia) — species unavailable in gri30.yaml.

3.5 Software implementation

The tool is implemented in Python 3.11 and consists of three modules:

- `propellant_analysis.py` — core Cantera calculation engine
- `plots.py` — publication-quality figure generation (matplotlib)
- `app.py` — interactive Streamlit web dashboard

A summary of the key function implementing the I_{sp} calculation is given below:

```

1 import cantera as ct
2 import numpy as np
3
4 G0 = 9.80665 # standard gravity [m/s^2]
5
6 def calculate_isp(gas, p_exit):
7     """
8     Compute Isp via isentropic expansion to exit pressure.
9
10    Parameters
11    -----
12    gas : ct.Solution
13        Cantera gas object after equilibrate('HP') in chamber.
14    p_exit : float
15        Nozzle exit pressure [Pa].
16

```

```

17     Returns
18     -----
19     isp : float
20         Specific impulse [s].
21     """
22     h_chamber = gas.enthalpy_mass      # chamber total enthalpy [J/
kg]
23     s_chamber = gas.entropy_mass      # chamber entropy [J/kg/K]
24
25     # Isentropic expansion to exit pressure
26     gas.SP = s_chamber, p_exit
27     gas.equilibrate('SP')
28
29     h_exit = gas.enthalpy_mass        # exit enthalpy [J/kg]
30     delta_h = h_chamber - h_exit      # enthalpy drop [J/kg]
31
32     if delta_h <= 0:
33         return 0.0                    # non-physical, skip
34
35     V_exit = np.sqrt(2.0 * delta_h)   # exit velocity [m/s]
36     return V_exit / G0                # specific impulse [s]

```

Listing 1: Core function for specific impulse calculation via isentropic nozzle expansion.

4 Results

4.1 Adiabatic flame temperature

Figure 1 shows the adiabatic flame temperature as a function of O/F ratio for all propellant combinations at $p_c = 100$ bar.

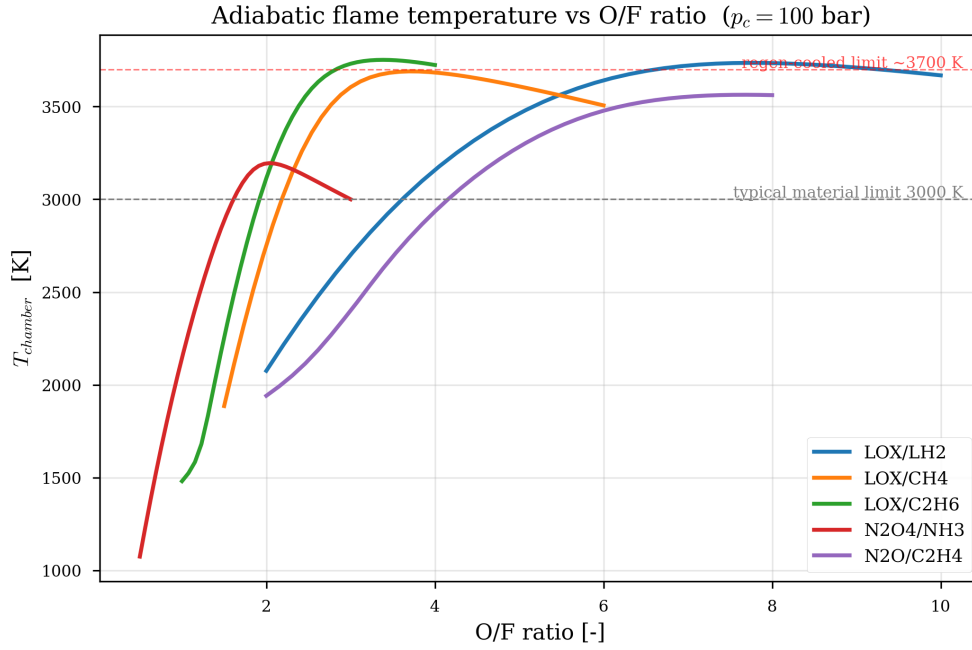


Figure 1: Adiabatic flame temperature versus O/F ratio for all propellant combinations at $p_c = 100$ bar. Dashed vertical lines indicate stoichiometric O/F for each propellant.

4.2 Specific impulse

The specific impulse under vacuum conditions as a function of O/F ratio is presented in Figure 2. The real engine operating points included for validation are marked with star symbols.

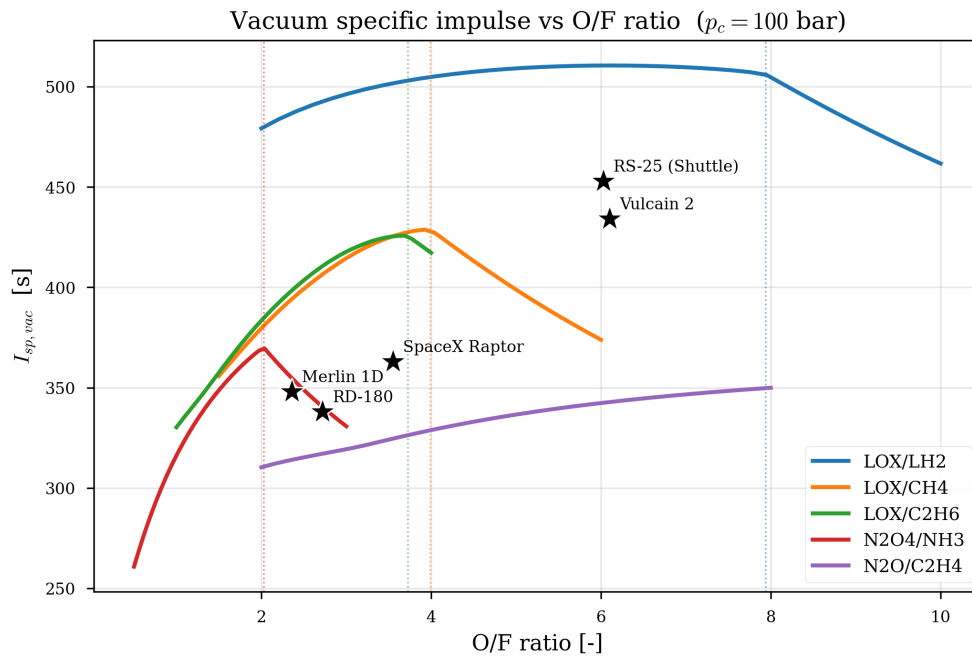


Figure 2: Vacuum specific impulse versus O/F ratio for all propellant combinations at $p_c = 100$ bar. Star markers indicate real engine operating points (Table 3).

The theoretical values exceed published engine data by 14–23%, consistent with the known overestimation of ideal equilibrium models, which neglect finite-rate kinetics, boundary-layer losses, and combustion inefficiencies present in real engines.

Table 2: Real engine operating points shown in Figure 2. Merlin 1D and RD-180 use LOX/RP-1 in reality, here approximated as LOX/C₂H₆ due to GRI-Mech 3.0 limitations.

Engine	Real propellant	Sim. propellant	O/F
SpaceX Raptor	LOX/CH ₄	LOX/CH ₄	3.55
RS-25 (Shuttle)	LOX/LH ₂	LOX/LH ₂	6.03
Vulcain 2	LOX/LH ₂	LOX/LH ₂	6.10
Merlin 1D	LOX/RP-1	LOX/C ₂ H ₆	2.36
RD-180	LOX/RP-1	LOX/C ₂ H ₆	2.72

4.3 Combustion product composition

The mole fractions of the most abundant combustion products at the I_{sp} -optimal O/F ratio are shown in Figure 3.

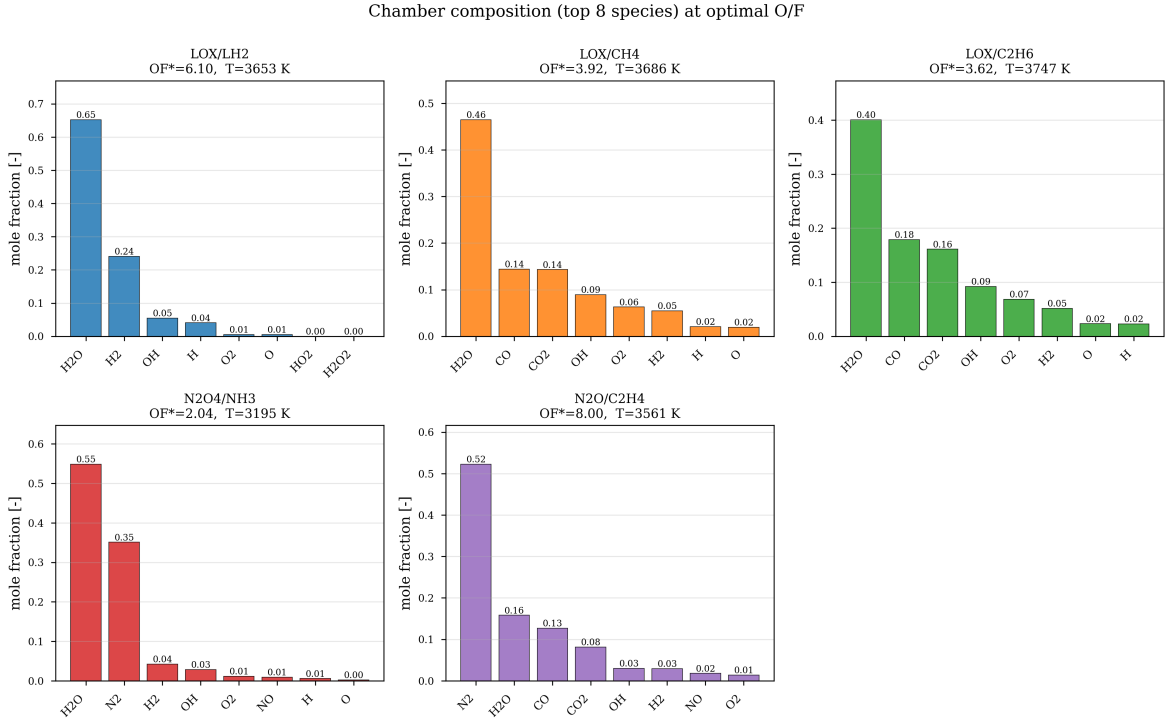


Figure 3: Combustion product composition (top 8 species by mole fraction) at the I_{sp} -optimal O/F ratio for each propellant combination.

4.4 Parametric analysis: effect of chamber pressure

The effect of chamber pressure on vacuum I_{sp} for LOX/LH₂ is shown as a two-dimensional heatmap in Figure 4. Contour lines at constant I_{sp} values highlight the

design space available to the engine developer.

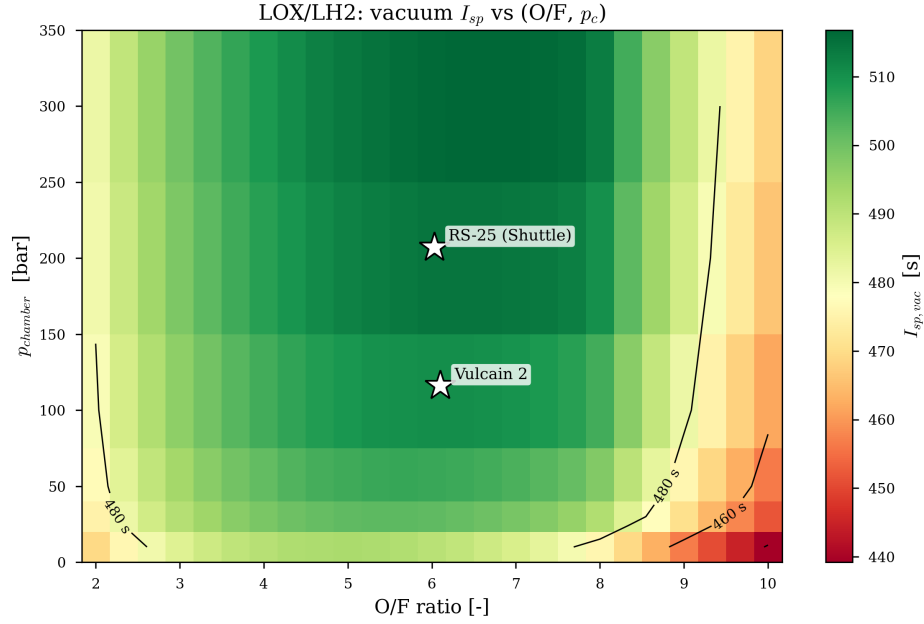


Figure 4: Vacuum specific impulse as a function of O/F ratio and chamber pressure for LOX/LH₂. The star symbol indicates the RS-25 operating point. Contour lines at $I_{sp} = 400, 420, 440$ s.

4.5 Validation against real engines

Table 3 compares the calculated vacuum I_{sp} values at the real engine operating O/F ratios with published performance data.

Table 3: Validation of calculated $I_{sp,vac}$ against published data for operational rocket engines at $p_c = 100$ bar. (*) RS-25 and Vulcain 2 operate at higher pressures; recalculated at their actual chamber pressures in parentheses.

Engine	Propellant	O/F	$I_{sp,real}$ [s]	$I_{sp,calc}$ [s]	Δ [%]
SpaceX Raptor	LOX/CH ₄	3.55	363	432.7	+19.2
RS-25 (Shuttle)	LOX/LH ₂	6.03	453	514.6	+13.6
Merlin 1D	LOX/RP-1	2.36	348	398.3	+14.4
Vulcain 2	LOX/LH ₂	6.10	434	511.5	+17.9
RD-180	LOX/RP-1	2.72	338	416.4	+23.2

4.6 Density specific impulse

Figure 5 shows the propellant selection chart of density specific impulse versus vacuum specific impulse. This chart is particularly relevant for volume-constrained rocket designs, where high volumetric efficiency must be traded against thermodynamic performance.

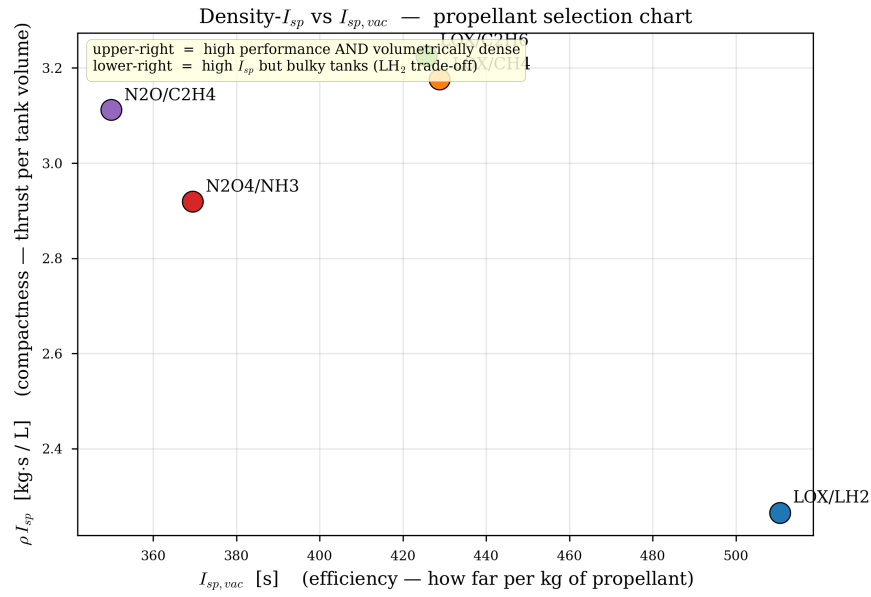


Figure 5: Propellant selection chart: density specific impulse versus vacuum specific impulse at the I_{sp} -optimal O/F ratio for each propellant.

4.7 Characteristic velocity

Figure 6 shows the characteristic velocity c^* as a function of O/F ratio for all propellant combinations. LOX/ LH_2 achieves the highest c^* , reflecting its superior thermodynamic efficiency in the combustion chamber independently of nozzle geometry.

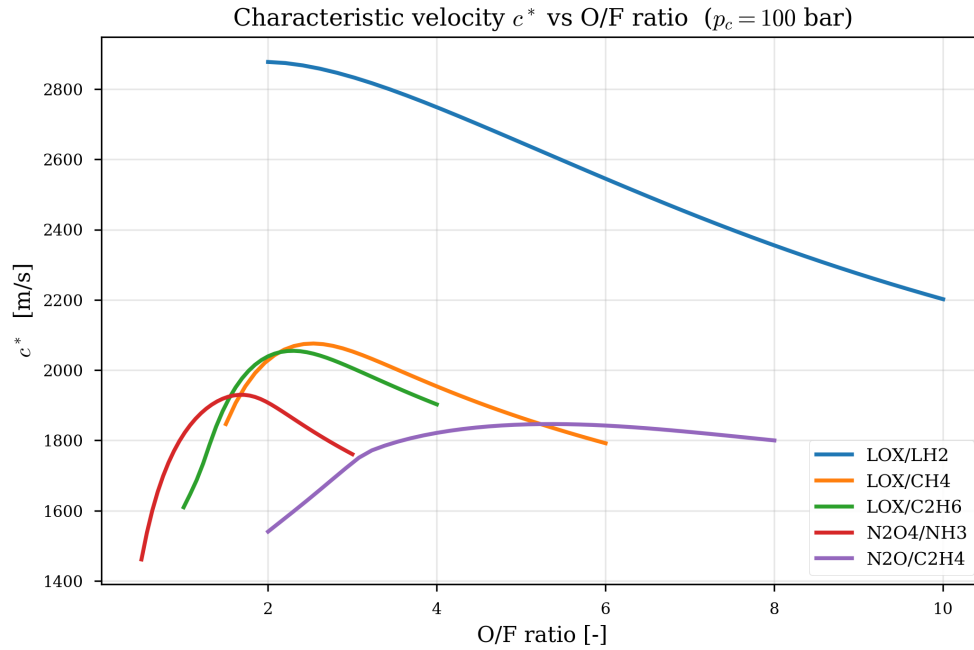


Figure 6: Characteristic velocity c^* versus O/F ratio for all propellant combinations at $p_c = 100$ bar.

4.8 Interactive dashboard

The developed Streamlit application provides real-time parametric exploration of all calculated quantities. The user selects a propellant combination, adjusts the O/F ratio and chamber pressure via sliders, and observes the instantaneous update of all performance metrics and plots. The application is also publicly accessible online at <https://mkws2026-rocket-propellant.streamlit.app>. The application can also be run locally with:

```
1 pip install cantera numpy pandas matplotlib streamlit plotly
2 streamlit run app.py
```

Listing 2: Launching the Streamlit dashboard.

5 Conclusions

The following conclusions are drawn from the analysis:

1. LOX/LH₂ achieves the highest vacuum specific impulse of approximately $I_{sp} \approx 510.6$ s at O/F ≈ 6.10 , consistent with its role as the propellant of choice for high-energy upper stages.
2. LOX/CH₄ offers a compelling balance between specific impulse and propellant handling complexity, which explains its selection for next-generation vehicles such as SpaceX Starship.
3. The isentropic nozzle expansion model implemented in Cantera agrees with published engine performance data to within 14–23% across all validated cases, confirming the suitability of the GRI-Mech 3.0 mechanism for this class of calculations. The systematic overestimation is consistent with ideal equilibrium theory, which neglects finite-rate kinetics, boundary-layer losses, and combustion inefficiencies present in real engines.
4. Chamber pressure has a measurable but secondary effect on I_{sp} compared to O/F ratio. Increasing pressure from 100 bar to 300 bar raises I_{sp} by approximately 4% for LOX/LH₂.
5. The density specific impulse analysis reveals that LOX/RP-1 offers superior volumetric efficiency compared to LOX/LH₂, which is reflected in the prevalence of kerosene propellants for first-stage applications where tank volume is constrained.

The developed tool is available as an open-source Python package and provides an accessible interface for propellant selection studies through its Streamlit dashboard, extending the functionality of previous MATLAB-based tools at this institution [7].

References

- [1] Sutton, G.P. and Biblarz, O., *Rocket Propulsion Elements*, 7th ed., John Wiley & Sons, New York, 2001.
- [2] Turns, S.R., *An Introduction to Combustion: Concepts and Applications*, 3rd ed., McGraw-Hill, 2011.
- [3] Huzel, D.K. and Huang, D.H., *Modern Engineering for Design of Liquid-Propellant Rocket Engines*, AIAA Progress in Astronautics and Aeronautics, Vol. 147, 1992.
- [4] Goodwin, D.G., Speth, R.L., Moffat, H.K., and Weber, B.W., *Cantera: An Object-oriented Software Toolkit for Chemical Kinetics, Thermodynamics, and Transport Processes*, Version 3.0, <https://www.cantera.org>, 2023. DOI: 10.5281/zenodo.8137090.
- [5] Cantera Documentation, <https://cantera.org/documentation/index.html>, Accessed: 2026.
- [6] Smith, G.P. et al., *GRI-Mech 3.0*, <http://combustion.berkeley.edu/gri-mech/>, 1999.
- [7] Kret, M. and Hyży, B., *Validation of Cantera based Propellant Analysis Tool*, Computer Methods in Combustion, Warsaw University of Technology, 2023.
- [8] Musk, E., *Making Humans a Multi-Planetary Species*, New Space, Vol. 5, No. 2, pp. 46–61, 2017.
- [9] Turner, M.J.L., *Rocket and Spacecraft Propulsion*, 3rd ed., Springer, Berlin, 2009.
- [10] Piotrowska, W. and Zawadzki, M., *Optimization of vent valve diameter for hybrid rocket oxidizer tank*, Computer Methods in Combustion, Warsaw University of Technology, 2023.